Autonomous Retail Researcher Agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Problem Statement & Objectives

1.1 Problem Statement

The retail industry operates in a fast-changing environment where market trends, customer preferences, competitor strategies, and pricing models evolve continuously. Retail organizations rely heavily on market research to make informed business decisions. However, traditional retail research methods are largely manual, time-consuming, costly, and dependent on human effort.

Manual research requires analysts to browse multiple websites, collect information from different sources, compare data, and then summarize findings. This approach is inefficient and not scalable, especially when real-time or frequent analysis is required. Additionally, human researchers may introduce bias, inconsistency, or delays in data processing.

With advancements in Artificial Intelligence and Large Language Models (LLMs), there is a strong need to automate retail research tasks using intelligent systems. The challenge is to design an autonomous system that can understand user queries, search for relevant retail information, analyze data, and generate structured insights with minimal human intervention.

This project addresses this challenge by developing an Autonomous Retail Researcher Agent using CrewAI, which coordinates multiple specialized AI agents to perform retail research tasks efficiently.

1.2 Project Objectives

The primary objectives of this project are as follows:

- To design and develop an autonomous AI-based system for retail market research
- To implement a multi-agent architecture using CrewAI
- To enable the system to understand natural language retail queries
- To automate web-based data collection and analysis
- To integrate real-time data with historical knowledge using Retrieval Augmented Generation (RAG)
- To generate structured and meaningful retail research reports
- To reduce manual effort and improve research efficiency
- To demonstrate the real-world application of LLM-powered autonomous agents.

1.3 Scope of the Project

The scope of the project includes:

- Automated research of retail market trends
- Competitor and pricing analysis
- Generation of summarized retail insights

- Use of multiple AI agents working collaboratively
- Dynamic data gathering from web sources

The project does not focus on predictive analytics, large-scale databases, or long-term enterprise deployment.

2. Proposed Solution

The Autonomous Retail Researcher Agent is a multi-agent AI system designed using CrewAI to coordinate multiple specialized agents for retail research tasks. The system follows an Agent + Tool + LLM pipeline, where each agent performs a well-defined role within the workflow.

The solution integrates:

- CrewAI for agent orchestration
- LLM API (Groq / LLaMA) for reasoning and text generation
- Web search tools for real-time retail research
- Python-based backend logic
- CLI or Streamlit interface for user interaction

Upon receiving a retail-related query, the system autonomously executes a collaborative research workflow and produces a structured retail report.

2.1 Key Features

The key features of the system include:

- Autonomous retail research execution
- Multi-agent collaboration (Researcher, Analyst, Price Analyst, Writer)
- LLM-powered contextual understanding and reasoning
- Real-time web data integration
- Automated report generation
- Modular and scalable agent-based architecture
- Minimal human intervention

2.2 Overall Architecture / Workflow

The proposed Autonomous Retail Intelligence System follows a Multi-Agent Retrieval-Augmented Generation (RAG) architecture designed to perform intelligent retail market analysis through coordinated AI agents and persistent semantic memory. The system is structured into multiple logical layers including the presentation layer, orchestration layer, intelligence layer, and memory layer. These components work together to enable automated research, competitive analysis, pricing evaluation, and professional business report generation.

At the presentation layer, the system uses Streamlit to provide an interactive web-based interface where users can enter retail market queries. This interface serves as the entry point of the system, capturing user input and displaying structured analytical reports generated by the AI agents. The interface triggers the backend workflow execution whenever a query is submitted.

The orchestration layer is implemented using CrewAI, which manages a sequence of specialized agents. The architecture includes a Research Agent responsible for collecting relevant market information, an Analyst Agent that performs competitive and strategic analysis, a Price Analyst Agent that evaluates pricing strategies and market positioning, and a Report Writer Agent that generates the final structured business report. These agents operate sequentially, passing context from one stage to the next to ensure logical consistency and structured reasoning throughout the workflow.

The intelligence layer is powered by the Llama 3 70B model accessed via the Groq API. This large language model provides advanced natural language understanding, analytical reasoning, and strategic content generation capabilities. Groq's high-speed inference infrastructure ensures efficient multi-agent execution with minimal latency, making the system responsive even for complex analytical queries.

To enhance contextual awareness and long-term memory, the system integrates a Retrieval-Augmented Generation mechanism supported by ChromaDB as a persistent vector database. Generated reports are converted into embeddings and stored in the vector database. When a new query is submitted, the system retrieves semantically similar past reports using vector similarity search. These retrieved insights are injected into the research stage, allowing the agents to reference historical knowledge and produce more informed, context-aware outputs. This creates a feedback loop where each new report strengthens the system's analytical memory.

The workflow proceeds as follows:

1. The user enters a retail-related query through the Streamlit-based user interface.
2. The Streamlit frontend forwards the query to the CrewAI Orchestrator for task coordination.
3. The orchestrator initializes a multi-agent workflow, assigning predefined roles to specialized AI agents.
4. The Researcher Agent collects relevant retail information using web search tools and stores useful findings in the knowledge base.
5. The Analyst Agent processes the collected data and performs market and competitor analysis.
6. The Pricing Agent evaluates pricing trends and strategic insights based on the analyzed data.

7. The Writer Agent consolidates all intermediate outputs and generates a structured retail research report using an LLM.
8. The final report is stored in the ChromaDB-based knowledge repository and returned to the user via the Streamlit interface.

This architecture ensures a clear separation between presentation, orchestration, intelligence, and data layers, while enabling autonomous, scalable, and collaborative reasoning through a multi-agent AI system.

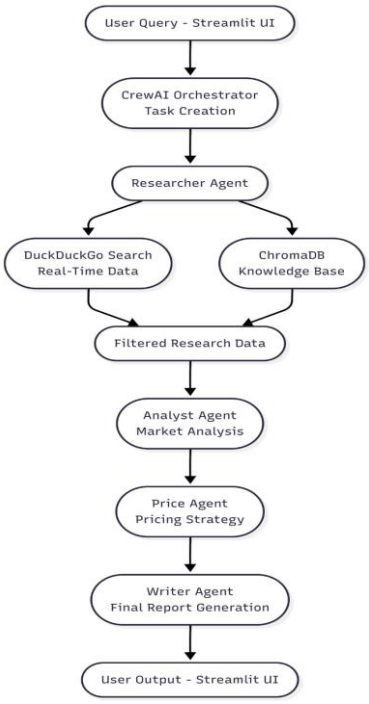


Fig. System Workflow Diagram of Autonomous Retail Researcher Agent

2.3 Tools & Technologies Used

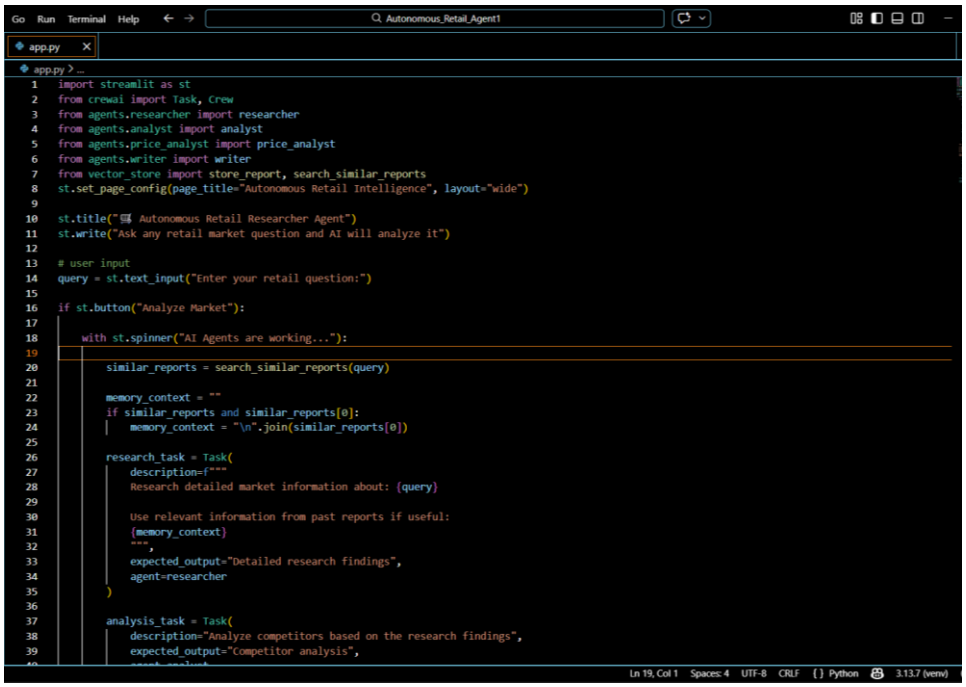
The system integrates a robust set of modern technologies to ensure efficiency, scalability, and intelligent automation in retail research.

Technology	Description
Python	Primary programming language used for developing the core logic of the system and integrating all components.

Streamlit	Used to build an interactive and user-friendly web interface for accepting queries and displaying results.
CrewAI	Framework used for orchestrating and coordinating multiple autonomous AI agents in a structured workflow.
Groq (LLaMA 3.3 70B)	Large Language Model used for natural language understanding, reasoning, and generation of retail insights.
ChromaDB	Vector database used for efficient similarity search and semantic retrieval in Retrieval-Augmented Generation (RAG).
DuckDuckGo Search API	Used to retrieve real-time, up-to-date retail market information and competitor data from the web to support accurate and current analysis.
Git	Use for version control ,source code management and collabration

3. Results & Output

3.1 Code Screenshots



```python
import streamlit as st
from crewai import Task, Crew
from agents.researcher import researcher
from agents.analyst import analyst
from agents.price_analyst import price_analyst
from agents.writer import writer
from vector_store import store_report, search_similar_reports
st.set_page_config(page_title="Autonomous Retail Intelligence", layout="wide")

st.title("🤖 Autonomous Retail Researcher Agent")
st.write("Ask any retail market question and AI will analyze it")

# user input
query = st.text_input("Enter your retail question:")

if st.button("Analyze Market"):

    with st.spinner("AI Agents are working..."):

        similar_reports = search_similar_reports(query)

        memory_context = ""
        if similar_reports and similar_reports[0]:
            memory_context = "\n".join(similar_reports[0])

        research_task = Task(
            description=f"""
            Research detailed market information about: {query}

            Use relevant information from past reports if useful:
            {memory_context}
            """,
            expected_output="Detailed research findings",
            agent=researcher
        )

        analysis_task = Task(
            description="Analyze competitors based on the research findings",
            expected_output="Competitor analysis",
```

Main Code

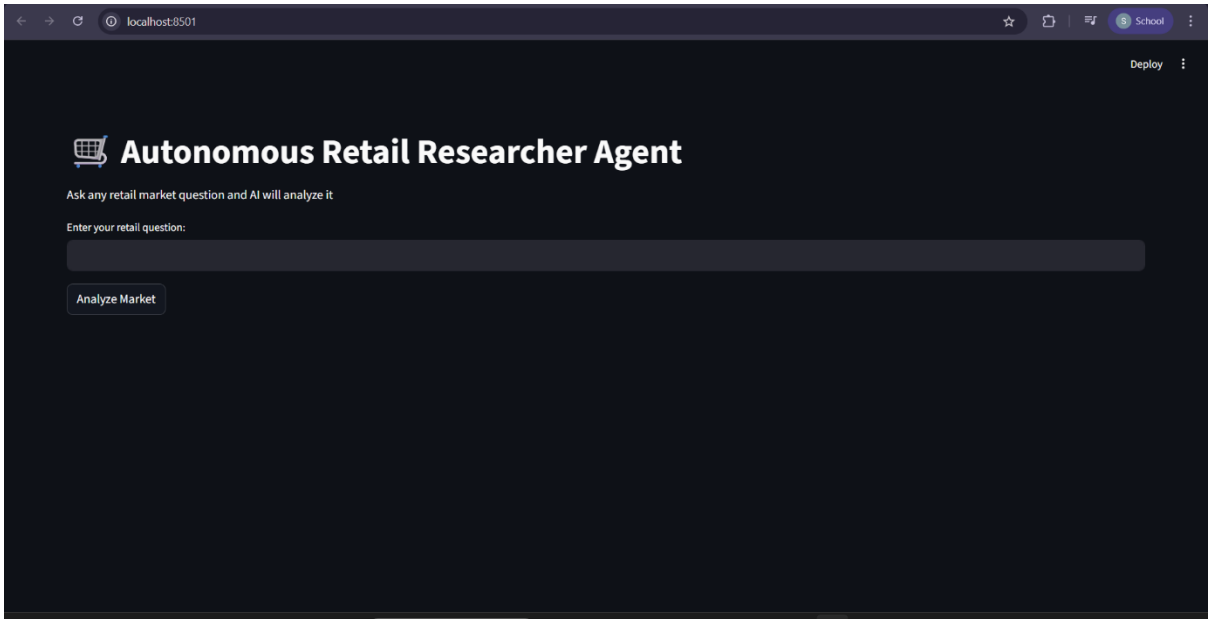
```python
from crewai import Agent, LLM
from dotenv import load_dotenv

load_dotenv()

llm = LLM(model="groq/llama-3.3-70b-versatile", temperature=0)

analyst = Agent(
    role="Retail Market Analyst",
    goal="Analyze competitors and market positions",
    backstory="Expert in understanding brand strategies, strengths and weaknesses in retail industry",
    llm=llm,
    verbose=True,
    allow_delegation=False,
    max_iter=2
)
```

Agents Code

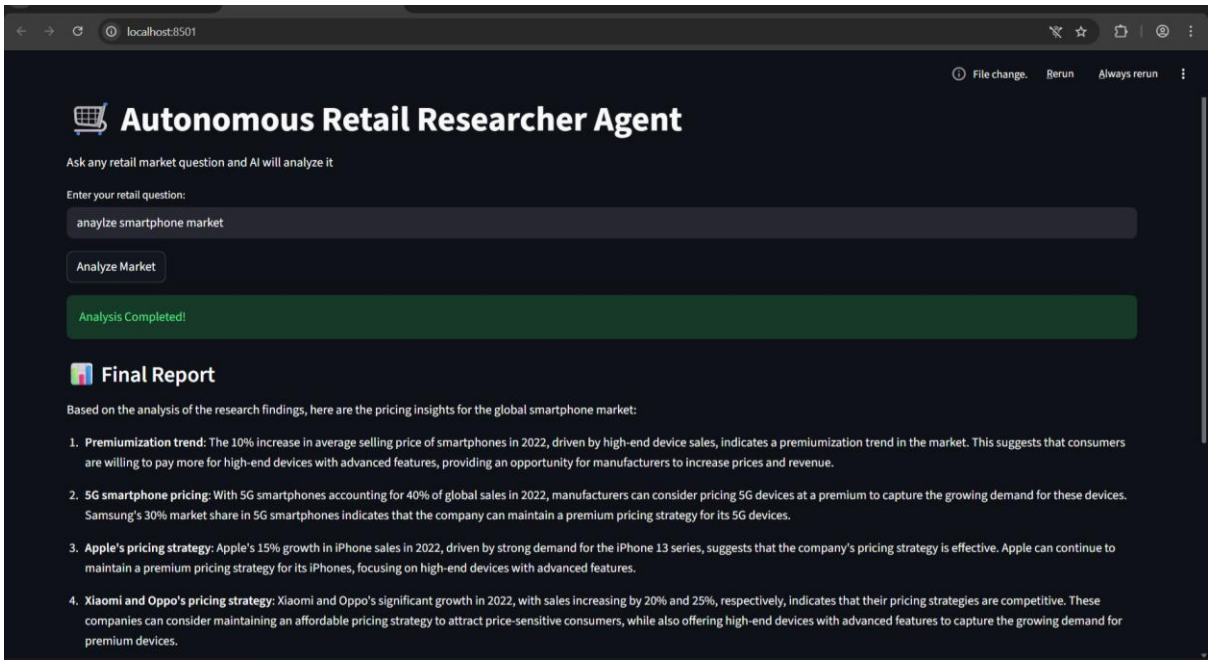
```python
import chromadb
from chromadb.utils import embedding_functions
from datetime import datetime


client = chromadb.PersistentClient(path="./chroma_storage")

embedding_function = embedding_functions.SentenceTransformerEmbeddingFunction(
    model_name="all-MiniLM-L6-v2"
)

collection = client.get_or_create_collection(
    name="retail_reports",
    embedding_function=embedding_function
)

def store_report(report_text):
    report_id = f"report_{datetime.now().timestamp()}"
    collection.add(
        documents=[str(report_text)],
        ids=[report_id]
    )

def search_similar_reports(query, n_results=2):
    results = collection.query(
        query_texts=[query],
        n_results=n_results
    )
    return results["documents"]
```

ChromaDB code

3.2 Output



User Interface



Final Report After Analysis

3.3 Key Outcomes

- Developed an Autonomous Multi-Agent Retail Intelligence System capable of end-to-end market analysis.
- Successfully implemented a Multi-Agent Workflow using CrewAI, enabling structured task delegation (Research → Analysis → Pricing → Reporting).
- Integrated high-performance reasoning using Llama 3 70B via the Groq API, ensuring fast and accurate analytical outputs.
- Implemented Retrieval-Augmented Generation (RAG) to enhance contextual intelligence and improve report quality.
- Integrated persistent vector memory using ChromaDB, enabling:
- Storage of past reports
- Semantic similarity search
- Cross-session knowledge retention
- Enabled generation of structured professional business reports, including key trends, pricing insights, and strategic recommendations.
- Reduced manual effort in retail market research by automating:
- Competitor analysis
- Market trend identification
- Pricing strategy formulation
- Built a scalable and modular architecture that allows future expansion (e.g., forecasting, demand modeling, financial projections).
- Designed an interactive Streamlit-based user interface for real-time retail intelligence analysis.
- Demonstrated practical application of AI, RAG, and vector databases in real-world business decision support systems.

4. Conclusion

The Autonomous Retail Intelligence System successfully demonstrates the practical implementation of a multi-agent AI architecture combined with Retrieval-Augmented Generation to solve real-world retail market analysis challenges. By integrating specialized agents for research, competitive analysis, pricing evaluation, and structured report generation, the system achieves a high level of automation and analytical coherence. The use of the Llama 3 70B model through the Groq API ensures fast and advanced reasoning capabilities, enabling the system to generate professional and data-driven insights efficiently. Furthermore, the integration of persistent semantic memory using ChromaDB enhances contextual understanding by allowing the system to reference past reports and continuously improve its analytical depth. Overall, the project highlights the effectiveness of combining multi-agent collaboration, large language models, and vector databases to build scalable, intelligent decision-support systems. This work demonstrates a significant step toward autonomous AI-powered business intelligence solutions capable of supporting strategic retail decision-making.

5. Future Scope & Enhancements

- Integration of real-time retail and e-commerce APIs for live market data analysis.
- Implementation of demand forecasting and sales prediction models using time-series analysis.
- Addition of new specialized agents such as:
- Customer Sentiment Analysis Agent
- Inventory Optimization Agent
- Financial Risk Assessment Agent
- Trend Forecasting Agent

- Fine-tuning or domain adaptation of Llama 3 70B for retail-specific datasets to improve analytical accuracy.
- Enhancement of vector search capabilities in ChromaDB using hybrid search and optimized indexing techniques.
- Integration of advanced data visualization dashboards (graphs, charts, competitor comparison tables).
- Cloud deployment for enterprise scalability and multi-user access.
- Implementation of role-based authentication and enhanced security mechanisms.
- Support for multi-language queries and global retail markets.
- Automated export of reports in PDF, Excel, and business presentation formats.
- Integration with ERP and CRM systems for enterprise-level adoption.